

Scale-Coupled Dynamics II: foundations re-examined, the pairing, and a torsion-free antisymmetric Ricci

Abstract

Scale-Coupled Dynamics (SCD) is a formalised framework in which the only primitive is a local scale and the geometry is its deformation. This note reports a revision pass carried out entirely inside the Lean 4 formalisation: thirty-four new source files and thirty-three new register sections. Its results fall into four groups.

Foundations. The structure carrying axioms A2 and A3 had *never been constructed* — every theorem about the scale field was true, clean and unwitnessed. Two model families are supplied. A periodic scale field is then shown to be impossible: if the log-scale is defined only modulo a period it is not a ring element at all, and only its gradient is. Measuring what A2 actually assumes shows the geometry never sees the log-scale, only its gradient; that A2’s content is *one unit*; and that what it silently adds is *exactness*, which is the branch of A3’ the framework’s own gravity sector does not use. The founding statement $scale \times energy = constant$ then needs only a group homomorphism, the commutative ring having been supplying a second, independent thing — a bilinear pairing for the geometry. Seven axioms become five.

What can be predicted. Predictions are informationally disjoint from inputs, and counts cannot fix a unit; the framework’s one dimensionless number is therefore a *permanent* input, both of its categories being closed. An input buys a prediction exactly when it appears twice. The escapes to real numbers, re-invented eight times with no theorem relating any two, are shown to be a single presheaf plus a *size*; the size adds no freedom, so “uniform density” is precisely the assumption that makes the size and the modulus one parameter.

The source and the pairing. No group-valued charge can be a universally attractive additive source, so the source of the geometry cannot be topological at all. Separately, the pairing was never chosen: the coupling was computed with a Euclidean trace while the observables live in a Lorentzian sector, joined by one theorem whose hypotheses are mutually exclusive. Redoing the curvature identity with a signature leaves $\kappa = -2(n-1)\Delta$ unchanged, for a structural reason, but the operator it multiplies is the d’Alembertian: the source law is hyperbolic and Poisson’s equation is its static limit. Scale disturbances then ride the framework’s own null cone, and their effect is quantised by the threshold spectrum.

Quantum. Off the commuting locus the Ricci tensor is *not symmetric*, with $Ric_{be} - Ric_{eb} = (n-2)[\sigma_b, \sigma_e]$, while the connection remains torsion-free and the scalar curvature does not move at all. Nothing classical takes a quantum correction at any order; in particular there is no $\hbar G/r^3$ correction to the Newtonian potential and no running of G . Finally the usual four-constants–three-dimensions bookkeeping collapses to one grading in which $\hbar$ and G carry the *same* weight, making “are they the same magnitude?” a well-formed question whose answer is one pure number.

Five claims of the development are withdrawn, and three are kept while their stated grounds are replaced. All statements are theorems in a Lean 4 development of 1275 theorems, every one of which passes `#print axioms` with no axiom beyond `propext`, `Classical.choice` and `Quot.sound`, and no `sorry`.

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1 Scope, and how to read this

SCD posits a dimensionless local log-scale σ over a ring with n commuting derivations, and derives geometry as the deformation of that scale rather than as an independent metric structure. This note does not re-derive the framework. It reports a single revision pass and is written so that every claim can be checked against a named Lean declaration.

Three words are used with fixed meanings, and they are not decoration.

- **Claimed** — proved in the formalisation, with the declaration named.
- **Identified** — an interpretive step the formalisation records as an input rather than a consequence.
- **Withdrawn** — a claim previously made in this development and now retracted, with the reason.

Section 22 collects what remains open, including one item this pass deliberately declined to pursue, and Section 23 reports the pass's own error rate, which is the part of a revision note usually omitted.

Part I

Foundations re-examined

2 A2 had no model, and now has two

An earlier audit had asked whether the framework's *auxiliary* structures are ever built from its own carrier. Nobody had asked the same question of the carriers themselves.

`ScaleField` — the structure carrying A2 and A3 — had **no term anywhere in the development**. Not “only in special cases”. Every theorem about the scale field was true, clean, and unwitnessed.

The reason is instructive: over a polynomial ring the only units are the constants, so $\partial\sigma = 0$ and the scale cannot vary. `ExpPoly.lean` constructs an exponential-polynomial ring in which e^p is a unit, giving the first non-constant scale field and the first anisotropic directional scale, with an exact reciprocal at finite amplitude.

Remark 2.1 (A second mechanism, and a scope statement). Exponentiation is not forced. `ExpPoly.dualScaleField` supplies a nilpotent model that the development had carried for other purposes without recognising it as a scale field, and `ExpPoly.exp_not_first_order` records the difference: nilpotent is exact but truncates at finite order. *Both* models sit on the $\mathbb{R}$ branch of A3' — and the framework's own analysis says gravity selects the other branch. So what is closed is “A2 has no model”, on the branch the source law does not use.

3 There is no periodic scale field

The natural next object — a model that is at once A1–A2 and periodic, so the two branches of A3' share a carrier — does not exist.

Theorem 3.1 (`Circle.no_scaleField_with_uPow`). *If the log-scale is defined only modulo a period then σ is **not an element of the ring**; only its gradient is, because differentiating kills the ambiguity. In $\mathbb{R}[u, u^{-1}]$ with $d = u \partial_u$ the unit u^k has a perfectly good scale gradient — the constant k — and for $k \neq 0$ no scale field carries it.*

So `Axioms.ScaleField`, which has a field $\sigma : A$, is the $\mathbb{R}$ branch, by construction rather than by modelling accident. The obstruction is exactness: a non-zero constant is not a derivative, so the gradient is closed and not exact.

Definition 3.2 (`Circle.ScaleGradient`). A unit s , a family grad , and $ds = s \cdot \text{grad}$. Exactness separates the two branches, and `Circle.exact_iff_trivial` connects their carriers.

4 What A2 assumes, measured

Theorem 4.1 (`Gradient.geometry_congr`). *Every geometric object — Hessian, Laplacian, squared gradient, connection, Riemann, Ricci, deformation — is constructed from the gradient alone, so two scale fields with the same gradient have the same geometry.*

The geometry never sees the log-scale. It sees only its gradient.

Theorem 4.2 (`Gradient.grad_unique`, `Gradient.grad_eq_logDeriv`). *Given a unit s , the equation $ds = s \cdot g$ determines g . A scale gradient therefore adds nothing to s : **A2's content is one unit**, the gradient is a definition, and the log-scale is an optional potential.*

A2 does not assume “there is a scale”; it assumes the scale's gradient is **exact** — the $\mathbb{R}$ branch of A3', chosen silently, and by the framework's own summary the branch gravity

does not select.

Remark 4.3 (A cost that was not a cost). An earlier statement of this result listed the Hessian’s symmetry as a casualty of the reduction. `Gradient.logDeriv_closed` disproves it: the gradient of *every* unit is closed, so the symmetry survives with no hypothesis. The claim is left in place with the correction beside it, because the shape of the mistake — assuming a condition must be assumed — is the one worth remembering.

Theorem 4.4 (`Torus.torGradient_not_isExact`). *Inexactness is not a one-dimensional artefact: the group algebra of $\mathbb{Z}^n$ supplies witnesses in every dimension, with $n = 2$ explicit.*

4.1 The amendment, weighed and declined

Restating the axioms with the gradient primitive would have made visible which results need the potential. `Amendment.lean` supplies that information directly instead.

Theorem 4.5 (`Amendment.same_unit_same_geometry`, `Amendment.potential_unique_up_to_fiducial`). *Same unit implies identical geometry: no geometric result needs the potential. And A5’s fiducial freedom is the choice of potential.*

The rewrite therefore buys only tidiness and is not made; the axiom text records the finding instead.

5 The founding statement needs a group, not a ring

Theorem 5.1 (`Reduction.logDeriv_mul`). $\log\text{Deriv}(st)_i = \log\text{Deriv}(s)_i + \log\text{Deriv}(t)_i$.

The log-derivative is a *group homomorphism* into the additive group of gradients. This had never been written.

Corollary 5.2 (`Reduction.logDeriv_inv`). $\log\text{Deriv}(s^{-1}) = -\log\text{Deriv}(s)$ — *which is A3. That the energy’s gradient is minus the scale’s is inversion in a group, not an independent postulate.*

Definition 5.3 (`Reduction.ScaleHom`). A group G , an abelian group M , and $D: G \rightarrow M^n$ with $D(st) = Ds + Dt$. Every scale algebra furnishes one. The signature contains *no ring, no commutativity, no multiplication*.

Theorem 5.4 (`Reduction.gradsq_add`). $\text{gradsq}(\sigma + \tau) = \text{gradsq } \sigma + \text{gradsq } \tau + 2 \sum_i \text{sig } \sigma_i \cdot \text{sig } \tau_i$.

The axiom is linear in the gradient; the geometry is quadratic; and the defect is exactly a *symmetric bilinear pairing*.

A commutative ring supplies the homomorphism and the pairing in one move. That is why it was chosen, and it was never recorded that these are two commitments rather than one. In a ring, the multiplication defining the unit group and the multiplication supplying the pairing are the *same operation*, and nothing in *scale $\times$ energy = constant* requires that.

6 Axiom consolidation: seven to five

Proposition 6.1 (Consolidation.A3_is_inversion). *The duality relation is Units.inv_mul and $\varepsilon := s^{-1}$ is a definition. A3 names a definition and asserts a group law.*

Theorem 6.2 (Consolidation.d_s_iff_exact). $(\forall i, \partial_i s = s \cdot \text{sig } \sigma_i) \iff (\forall i, \text{logDeriv}(s)_i = \text{sig } \sigma_i)$.

A2's derivation condition holds exactly when the log-derivative is *exact*; A4' already supplies the unit; and §4 showed no geometry needs the potential. The consolidated list is

$$\text{A1} \cdot \text{A2}^* (\text{one family of units}) \cdot \text{A5} \cdot \text{A6} \cdot \text{A6}'' \cdot \text{A7}.$$

The reduction is not claimed to be *forced*; what has changed is that the potential is now known to be the only surplus.

Part II

The observer, resolution, and what can be predicted

7 Operator ordering, and the observer screen

The quantitative gravity chain computes over a commutative ring, and carrying its formulas to a non-commutative ring requires an ordering the commutative computation does not determine.

Theorem 7.1 (Ordering.offDiag_sub_rev, Ordering.vacuum_is_ordering_dependent). *The two available orders of the off-diagonal Ricci summand differ by three commutators of scale gradients per transverse direction — the same objects the non-commutative sector calls the correction — and the difference reaches the vacuum condition.*

Theorem 7.2 (Ordering.ordering_irrelevant_of_static). *Under staticity the difference vanishes, so $\gamma = 1$, the PPN relation and the arcsecond figures are unaffected.*

The observer half of the same question:

Theorem 7.3 (Screen.observer_preserves_predictions). *A quantity constant across admissible worlds is constant across admissible (world, observer) pairs, because the pairs project into the worlds. No quantity of the world loses its status when the observer joins the configuration.*

Theorem 7.4 (Screen.observer_relative_is_modulus). *For every theory admitting one particular resolution — no constraint on the world assumed beyond that — what is recorded is a modulus: an input, not a prediction.*

8 Resolution without an order

The standing complaint was that the observer never touches the algebra, because the observer compares real values while a ring has no order. The diagnosis was too local: the emergence relation compares reals as well, so *resolution* is what had to change, not *observer*.

Definition 8.1 (`Resolution.Filtration`). A decreasing family of two-sided ideals; `mem k a` says a is invisible at resolution k , and `Resolved` is its negation. Monotonicity is the emergence result with the comparison removed.

Incompatibility is the framework’s own ad read at a resolution: `Compatible k a b := mem k (ad a b)`. No projection lattice is imported — idempotents had been declined precisely because adopting them imports the answer.

Theorem 8.2 (`Layers.compat_zero`, `Layers.incompat_one`). *Filtering 2×2 matrices by powers of a scalar matrix gives a pair compatible at resolution 0 and incompatible at 1: **the resolution index does work**, where before it was decoration.*

Theorem 8.3 (`Layers.resolved_is_emergence_resolved`, `Layers.monomial_threshold`). *The two notions of “resolved” are one, and the thresholds are computed from the ring — X^j has threshold j , its degree — where the emergence account had taken them as free data.*

An outcome is then an indistinguishability class (`Layers.indist_equivalence`). What is missing is its *weight*: a probability is a measure on classes, and the framework’s only native measure is counting.

9 Observer and observed are one type

Theorem 9.1 (`Vantage.observer_and_observed_are_one_type`). *Resolution and Threshold are both $\text{Fin } n \rightarrow \mathbb{R}$ — each is “one scale per direction”, which is $A4'$. The proof is `rfl`; the statement is the content.*

`Vantage.every_vantage_observs_itself` and `Vantage.self_observation_is_exact` are reflexivity and antisymmetry, so the vantage space is **a poset whose order is observation**.

The framework has no position space — its own axiom text says directions are indices at a point, not positions — so “the relation between the observer and spacetime” has no second term. What exists is a vantage space, and an observer is a point of it together with a commitment.

Content is *covariant* in the vantage (`Vantage.record_mono_in_resolution`, new here) and *contravariant* in the commitment, so it is a bifunctor (`Vantage.the_two_variances_commute`). An observer committing no direction records everything; committing all of them is detachment, and it records least — an uncertainty-shaped statement obtained by unfolding an intersection, and not claimed to be more.

Theorem 9.2 (`Vantage.algebra_record_is_observer_record`). *Given a filtration per direction, `Vantage.place` sends a ring element to one real per direction — its threshold there — and the algebraic resolved set is the observer’s record. This is the first map from the algebra into the observer’s space.*

The cost is stated: `place` needs n filtrations, and a filtration is postulated, so this multiplies an unwitnessed structure rather than discharging it.

10 What can be predicted, and what cannot

10.1 Equal spacing dies against the data

A filtration by powers of one element spaces its thresholds evenly.

Theorem 10.1 (`Spacing.thresholds_not_power_spaced`). *The measured spectrum's gaps are not evenly spaced, so no such filtration reproduces them.*

`Spacing.gapList_shift_invariant` and `Spacing.gapList_rescale` record what the gap list does and does not see, and `Spacing.unique_pure_number` isolates the one number that survives.

10.2 Predictions carry no information about inputs

Theorem 10.2 (`Information.predicted_cannot_determine_modulus`). *If f is constant across a theory's solutions and g is not, then g is not a function of f : the two solutions that disagree about g agree about f .*

Trivial as a proof and not as a statement — it retires an entire class of routes. A prediction can never fix a free number.

10.3 And counts cannot fix a unit

Theorem 10.3 (`Conversion.counts_cannot_fix_a_unit`, `Conversion.threshold_not_a_function_of_charge`). *A count is invariant under the rescaling that moves the unit, so no count determines it.*

$\Delta\rho$ is not an open problem. It is a **permanent input** — unless a quantity that is neither a prediction nor a count is introduced, and the framework has no third category.

10.4 But an input buys a prediction when it appears twice

Theorem 10.4 (`Overdetermination.one_number_two_relations`). *A free number occurring in one relation is absorbed by it and predicts nothing; in two, it is fixed by one measurement and predicts the other.*

The question therefore becomes a *count of appearances*, and `Overdetermination.coupling_from_rate_and_de` exhibits the shape a second appearance would have to take.

10.5 The rate's weight

Theorem 10.5 (`RateWeight.gauge_conversion_moves`, `RateWeight.prediction_at_unit_factor`). *The dissipation rate is **weight zero**, by two independent computations that agree.*

This kills one branch of the remaining dimensionful question and makes the gap exact.

11 The cosmological sector shares no types with the algebra

The cosmological files import only Mathlib; the log-scale does not occur in them; the scanning account takes its coupling, density and time as free real parameters and mentions the scale algebra nowhere — while the file where $\kappa = -2(n-1)\Delta$ lives does import the

algebra.

`Disconnect.factorisation_is_generic` and `Disconnect.factorisation_has_failing_instances` make it precise. There is no unit conversion to compute: the cosmological time variable is not the scale axis in other units, it is an unconnected real variable. *The bookkeeping is not hard — it is not yet a question.*

Remark 11.1 (Withdrawn). An earlier claim of this pass — that a bridge construction makes the scanning hypothesis a *theorem* of the index law (`Bridge.factorsThroughCount_of_indexResponse`) — is withdrawn. The theorem stands at the pair the bridge constructs; what was written around it, identifying that pair with the cosmological one, does not. `Observations.lean` keeps the two remaining identifications as explicit hypotheses carried visibly downstream. That also caught a trap: an identification written as a *structure* with the observed number as a field is trivially inhabited, since one can always name a real equal to a formula. It is written as a `Prop` parameterised by the observed number instead.

12 The unstated primitive: presheaves, and a size

Counting what the disconnection is a case of: at the time of the audit, **59 of 107 files never mentioned the scale algebra** (now 66 of 125), and the escape to real numbers had been re-invented **eight times**, with *no theorem relating any two*.

Theorem 12.1 (`Content.pullback_eq_emergence`). *Three of the framework’s own monotonicity results are functor laws. The algebraic sector and the real-valued threshold sector are **one presheaf described twice**; and the observer is a presheaf of the opposite variance.*

Theorem 12.2 (`Size.emergence_is_ofThreshold`, `Size.uniform_density_is_a_restriction`). *Thresholds and content are two descriptions of one datum, so the count of independent escapes was too high. What the rest are is not content but a size — a measure of content — and its linearity is a restriction, not a shape it must have.*

G4’s missing weight and the cosmological disconnection are **one gap**, approached from two sides, and it is the *size*.

Theorem 12.3 (`Valuation.observer_counting_law`, `Valuation.modularity_is_a_restriction`). *The framework’s own union law for observers, fed to a modular size, yields inclusion–exclusion on the observer lattice. Nothing picks the weight; nothing resists it either. Monotonicity alone is far too little — “one if anything is there” is monotone.*

12.1 And the size adds no freedom

Theorem 12.4 (`OneParameter.affine_size_determined`, `OneParameter.origin_is_a_fiducial_shift`, `OneParameter.size_adds_no_freedom`). *A uniform size is $\rho(t - t_0)$; that is all of it; its origin is A5’s rather than an observable; and its slope is the density already counted.*

The size adds no freedom. *Uniform density is not an assumption beside $\Delta\rho$: it is the assumption that makes them one parameter.*

The register had carried uniform density among its assumed items from the beginning without saying what it buys. This is what it buys.

Part III

The source and the pairing

13 The source of the geometry cannot be a topological charge

A6'' reads $\Delta\sigma = \Delta \cdot \nu$ and was presented as an *index* law with ν topological. Three candidate integers had been tabulated.

Theorem 13.1 (`Codimension2.two_valued_source_is_trivial`). *An additive map from a two-element group to $\mathbb{R}$ is identically zero, since $a^2 = 1$ forces $2f(a) = 0$. The projective order parameter's line defects carry exactly that charge, and `Codimension2.odd_pairs_cancel` is the framework's own selection rule.*

Theorem 13.2 (`DegreeSource.additive_nonneg_on_group_is_trivial`). *On a group, an additive $f: G \rightarrow \mathbb{R}$ with $f \geq 0$ is identically zero: $f(1) = 0$, then $f(a) + f(a^{-1}) = 0$ with both terms non-negative.*

One theorem covers both branches — the two-valued charge dies at $a = a^{-1}$, the signed degree at $f(k) + f(-k) = 0$ — and it never needed to know the codimension. Additivity is what makes a source an index; non-negativity is what universal attraction requires.

No group-valued charge can be a universally attractive additive source. Every topological charge here is group-valued, so the source of the geometry cannot be topological at all. The question “which codimension?” was malformed on both branches: it has an answer, and the answer is neither.

The result is sharp rather than over-strong. `DegreeSource.monoid_source_exists` gives an additive, non-negative, non-zero map on $\mathbb{N}$: it is *inverses* that do the killing, and a threshold count survives because content is a cardinality with no anti-content. Two external checks: real gravity's source is not topological and mass is non-negative; and electric charge *is* signed and *does* source an interaction, which the theorem permits because that interaction is not universally attractive.

Remark 13.3 (And the reproduced coefficient means nothing). `DegreeSource.degree_gives_the_same_kappa` is the original derivation *applied*, not *reproved*: its Δ was already an arbitrary ring element and the derivation used no property of it, so any constant gives the same formula with itself in the slot. “Does the degree version give the same κ ?” was a malformed question. What is touched is A6''’s standing: its authority came from ν being a topological index. A6'' *keeps its consequences and loses its index reading*.

14 The pairing was never chosen: two metrics

- `Conformal.RscBare` = $\sum_b \text{Ric}_{bb}$, called in its own documentation the δ -trace of Ricci, on the metric $e^{2\sigma}\delta$. This is where $\kappa = -2(n-1)\Delta$ is derived, so κ is a **Euclidean result**.

- `Diagonal.Ric` on a metric carrying the signature as a parameter with $\eta^2 = 1$. The development’s own chain table runs the *entire* gravity chain through this sector, and the key cancellation requires $\eta_t \eta_r = -1$.

Proposition 14.1 (`TwoMetrics.euclidean_excludes_lorentzian`). *If $2 \neq 0$, then $\eta \equiv 1$ and $\eta_t \eta_r = -1$ cannot both hold — and $\eta \equiv 1$ is the hypothesis of the one theorem joining the two sectors.*

The only bridge between the sectors holds exactly where the physics does not. κ and the observables have never been shown to be about the same geometry.

`TwoMetrics.delta_trace_ne_eta_trace`: on a Lorentzian pair the two traces read 2 and 0, so the difference does not cancel.

Remark 14.2 (A correction to the record). The framework’s summary of what rests on the axioms alone listed “the Lorentzian signature”. What is derived is a *codimension-one splitting* of the directions from the drift. A splitting is a flag; a signature is a property of a form — and that file’s own closing paragraph already granted that the timelike sign is not derived.

15 The repair: the coefficient survives, the operator does not

Theorem 15.1 (`EtaTrace.RscE_eq`).

$$R^\eta = -2(n-1)\square\sigma - (n-1)(n-2)|\nabla\sigma|_\eta^2,$$

with $\square\sigma = \sum_i \eta_i \sigma_{ii}$ and $|\nabla\sigma|_\eta^2 = \sum_i \eta_i \sigma_i \sigma_i$.

Identical coefficients, and `EtaTrace.A6’_from_index_eta` returns $\kappa = -2(n-1)\Delta$. The computation was redone from the connection upward, since inserting η into the final trace alone would not be the same geometry; the connection used is the one the development already had for the anisotropic sector, at the isotropic locus with η left general — a case that existed and had never been computed.

The reason is structural, not fortunate: in the two-connection contraction two η -carrying terms cancel identically, the second carrying $\eta_e^2 = 1$, and every surviving η is either squared or pinned by a δ that forces its indices equal. *The signature cannot reach the coefficient.*

Theorem 15.2 (`Hyperbolic.lapE_lorentzian_split`). *With $\eta_t = -1$ and $\eta_i = 1$ otherwise, $\square\sigma = \sum_{i \neq t} \sigma_{ii} - \sigma_{tt}$.*

κ is safe; the operator it multiplies is the d’Alembertian. The source law $\square\sigma = \Delta \cdot \nu$ is **hyperbolic** — a sourced wave equation — and Poisson’s equation is its *static limit*, which is where all the framework’s gravity results are computed.

Consequently the Poisson-limit theorem should be read as scoped to static configurations. Nothing computed from it changes; what changes is what it is a limit of.

Remark 15.3 (The bridge, half repaired). `Hyperbolic.diagonal_chr_isotropic` and `Hyperbolic.diagonal_ric_isotropic` join the two sectors at *any* signature, removing one of the joining theorem’s two hypotheses. The other — isotropy — survives, and the physics violates it, since the vacuum solution has $\sigma_t \neq \sigma_r$. The disconnect narrows from two obstructions to one.

16 Null characteristics, and what a scale wave is

A wave profile is a scale whose Hessian is rank one along a covector k ; no analysis is used, because none is available.

Theorem 16.1 (`Hyperbolic.vacuum_wave_is_null`, `Hyperbolic.characteristic_is_null`). $\square\sigma = \text{bareForm}(\eta, k) \cdot m$; in vacuum with m a unit the bare form vanishes; and that is the framework’s own null condition for any unit scale. With a Lorentzian signature it reads $\sum_{i \neq t} k_i^2 = k_t^2$.

Scale disturbances propagate on the framework’s own null cone. That cone had been defined *kinematically*; nothing had said anything travels along it.

Theorem 16.2 (`Directional.resolved_eq_iff_no_threshold_between`). Two scale values resolve the same content exactly when no threshold lies between them.

A scale wave has no continuous observable. It acts only where it crosses a threshold, so half the amplitude gives *none* of the effect unless a threshold lies in the interval. A scale wave is a wave of emergence.

16.1 The breathing mode: expressible, but not radiable

The development excluded a breathing polarization on the ground that a uniform rescaling is a fiducial shift.

Proposition 16.3 (`Breathing.nonconstant_scale_is_expressible`, `Breathing.uniform_across_directions_st`). Fiducial invariance quantifies over one constant added at every point, and the difference pattern compares points. A breathing wave is uniform across directions and varies across points, and a fiducial-invariant functional separates it. **A5 does not remove the breathing mode.**

The conclusion nevertheless survives on a ground the framework already had: a scalar disturbance is sourced by the monopole; a conserved source’s monopole is frozen; and a non-conserved source admits *no field equation at all*.

The breathing mode is expressible but cannot be radiated by any source for which a field equation can be written. Same prediction, different ground, narrower scope — a free or primordial breathing wave is excluded by nothing here.

Remark 16.4 (Withdrawn). An earlier statement of this pass — *no interferometer can see a scale wave* — is withdrawn. The cited theorem takes a *single* scale pattern and compares two null directions at one point, so it excludes finding a *static* anisotropy; a wave differs between the arms and between emission and return. What survives: a static scale pattern is invisible to an interferometer at any precision.

17 The directional source law

Theorem 17.1 (`Directional.dirTrace_factors_iff`). The conformal factor comes out of the metric trace only when every directional unit agrees.

So there is no “bare” scalar curvature off the isotropic locus, because there is no single conformal factor — and that is a property of A6′, whose form $e^{2\sigma}R = \kappa\rho$ presupposes an isotropic scale while A4′ supplies a directional one. The framework’s index response takes a *scalar* σ , and no source law for a directional scale exists anywhere. That absence is not an oversight.

Definition 17.2 (Breathing.DirSource). $\sum_b \eta_b w_b^{-1} R_{bb} = \kappa\rho$ — the law on the *true* scalar curvature.

A4′ supplies exactly the invertibility A6′’s bare form was contorted to avoid: its w are *units*. The consolidation the development wanted arrives in the directional sector, not the isotropic one.

Breathing.dirSource_isotropic reduces it to A6′ with the density differing by the conformal factor — coordinate against proper density, which A6′ could absorb and this cannot — and Breathing.kappa_from_dirSource recovers κ where the density cancels.

Part IV

The quantum sector

18 The correction is exact, and had never been carried

Theorem 18.1 (Micro.quantum_correction_signature_independent). *The signature enters the deformation tensor only symmetrically, so the antisymmetric part — which the development identifies as the whole quantum-gravitational correction — is the same for every η .*

Unlike κ , the quantum correction never depended on the pairing that was never chosen; Micro.DefmE_sub_Defm isolates where the pairing does matter, namely the symmetric part the classical tests read.

And NCConformal.quantum_correction_is_a_wedge holds by rfl: the correction *is* the framework’s rotational label applied to the scale gradient pattern — not analogous to it, the same term.

The chain files all carry CommRing, where the adjoint action vanishes identically. The correction is identified exactly and has **never been carried into a single computed number**. So “the quantum correction does not affect the classical tests” is at present as much a statement about *where the computation was done* as about the term’s symmetry.

Remark 18.2 (A near miss worth recording). The obvious citation here — that the scale and commutator channels cannot constrain each other — is *superseded*, as that file’s own header says: the independence is that of the *scalar* channel, and directional scalings bracket into a rotation, so under A4′ the channels are coupled. The correction being the wedge is that coupling seen again.

19 Off the commuting locus: an antisymmetric Ricci without torsion

Proposition 19.1 (QuantumRicci.sum_RmCorr_contract). *The Riemann correction vanishes in the Ricci contraction: its two Kronecker terms cancel term by term.*

Theorem 19.2 (QuantumRicci.two_Ric_eq_nc). $2\text{Ric}_{be} = (2 - n)D_{be} - \delta_{be} \text{tr } D$, with no extra term.

Yet the deformation tensor is itself not symmetric off the locus:

Theorem 19.3 (QuantumRicci.two_Ric_antisymm).

$$\text{Ric}_{be} - \text{Ric}_{eb} = (n - 2) [\sigma_b, \sigma_e]$$

Theorem 19.4 (QuantumRicci.two_RscBare_eq_nc). $2R = (2 - 2n) \text{tr } D$, and $\text{tr } D$ is built from σ_{aa} , $\sigma_a \sigma_a$ and $|\nabla \sigma|^2$ — every term a square.

The entire quantum correction to the geometry is the antisymmetric part of Ricci. The scalar curvature does not move at all.

19.1 What this predicts

An antisymmetric Ricci with a torsion-free connection. The connection is symmetric in its lower pair whether or not values commute (Conformal.Chr_symm, over a general ring), so there is *no torsion*. In ordinary differential geometry $\text{Ric}_{[ab]} = 0$ for a torsion-free connection, by the first Bianchi identity — whose proof uses commutativity of the *values*, not only torsion-freeness. This is therefore not Einstein–Cartan, which buys the same antisymmetry by giving the connection torsion.

Nothing classical moves, at any order. The source law, κ , the Newtonian limit and every cosmological statement are exactly their commutative selves. In particular there is **no $\hbar G/r^3$ correction to the Newtonian potential**, of the kind effective-field-theory treatments produce, and **no running of G** . This is a flat disagreement with the standard expectation, and it is exact rather than small.

It vanishes in two dimensions, by the same $(n - 2)$ that governs conformal geometry throughout.

Where it can appear. An antisymmetric Ricci in a field equation requires an antisymmetric source, and the physical one is spin. *This is an identification, not a theorem*: no field equation for the antisymmetric half is written, and no magnitude is claimed — calling the commutator an $\hbar$ -order effect runs through a separate registered identification. Nor is existence claimed: no scale algebra is exhibited carrying non-commuting gradients and a non-trivial geometry at once.

Usual	Here
L, T, M	one grading: weight under $\sigma \mapsto c\sigma$
c	not a constant — one cone; measured speed one is a theorem
$\hbar$	a weight-one scale step; an identification, not a construction
G	a weight-one coefficient Δ
e , i.e. α	weight zero — already pure, and structurally so
Planck units	one axis, so one magnitude fixes everything
pure numbers	$\Delta\rho$ and $\hbar\rho$: <i>two</i> , collapsing iff $\Delta = \hbar$

Table 1: The dimensional bookkeeping, translated.

Part V

Constants, open questions, method

20 Constants and dimensions

There are no dimensions here: what replaces dimension is a single $\mathbb{Z}$ -grading, the weight under $\sigma \mapsto c\sigma$. The collapse is structural. Time is a *direction*, told from space by a sign; light crosses at measured speed one in every direction because the scale sets rods and light times alike, so $c = 1$ is a theorem; and A3, being a group law (§6), places mass on the same axis with the opposite sign.

Theorem 20.1 (`Constants.two_pure_numbers`, `Constants.collapse_iff`). *Δ and the scale step identified with $\hbar$ both carry weight 1, and the threshold density weight -1 . So $\Delta\rho$ and $\hbar\rho$ are both weight zero, and they coincide if and only if $\Delta = \hbar$.*

There are *two* free pure numbers, not one, unless G and $\hbar$ are the same magnitude — and nothing in the development says they are.

The framework’s own files record that “ G and $\hbar$ share an origin” is conditional on a step not taken, and that “ $\hbar$ is that step” is an identification rather than a construction. The two statements had never been read against each other.

20.1 Where e goes

The fine-structure constant is weight zero — a gauge coupling’s value does not depend on how σ is labelled — so it is a pure number *structurally*. And the asymmetry has a reason: the gauge law relates weight-zero quantities to weight-zero quantities, while gravity’s relates a weight-zero *count* to a weight-one *scale*.

Exactly one law crosses the grading. “Why is α dimensionless and G not” stops being a fact about dimensions and becomes a fact about which law crosses the grading.

Remark 20.2 (A question the usual picture cannot ask). $[G] = L^3 M^{-1} T^{-2}$ and $[\hbar] = M L^2 T^{-1}$ have different dimensions, so *are G and $\hbar$ the same magnitude?* is not well-formed there. Here they carry the same weight, the question is well-formed, and its answer is a single pure number.

21 The prediction ledger

Parameter-free, and sharper than general relativity rather than different from it. $\gamma = 1$ exactly; the deflection ratio 2; no dipole radiation, from Bianchi rather than assumed, which binary-pulsar timing measures at 0.9983 ± 0.0016 ; *exactly zero* energy-dependent photon speed, where minimum-length approaches predict something small; *one cone for every sector*, so inter-sector Lorentz violation is inexpressible rather than bounded; entropy $\propto \ln R$; no horizon. All weight zero — nothing to tune.

New here, and mostly negative. A hyperbolic source law; null-cone propagation; a breathing mode expressible but not radiable; an effect quantised by the threshold spectrum; and an antisymmetric Ricci with no classical consequence at any order.

One number that cannot be computed. $\Delta\rho$ is a permanent input, so the framework predicts *ratios and structures, not magnitudes*. The familiar arcsecond figures are weight-zero statements evaluated on measured inputs.

Where the framework touches reality it mostly agrees with general relativity *while claiming more sharply* — exactly zero rather than small, inexpressible rather than bounded. That is real contact and it is testable. It is not yet a place where an experiment anyone has performed could tell the two apart.

22 What remains open

The directional curvature identity. `Breathing.DirSource` supplies the law; the identity that would let it produce a number does not exist off the isotropic locus. Until it does, κ and the observables meet only where the scale is isotropic. This is the single largest remaining gap, and it is a bounded, stateable computation.

The quantum correction has never been carried. §19 is the first computation off the commuting locus. Every number in the development is computed on the isotropic, commuting locus, and both unfinished sectors are off it.

The filtration is postulated. Nothing in the axioms builds one, and `Vantage.place` needs n of them.

$\Delta = \hbar$: **deliberately left open.** Nothing in the axioms relates the source coefficient to a scale shift, and the scale shift is not built from the axioms at all — it requires a ring endomorphism and a unit that nothing supplies. The identification would have to come from outside. The free-number count therefore reads *two*.

And the remainder. The non-static operator ordering; A7; the second-order content of the static vacuum equation; the absence of any site or covering; and G4's weight.

23 Method, and this pass's own error rate

Five claims of the development are withdrawn in this pass, and three are kept while their stated grounds are replaced: A6'' keeps its consequences and loses its index reading; the Poisson limit keeps its numbers and becomes a static limit; the breathing exclusion keeps its conclusion and changes its argument. Twice a file's own self-correction caught an error that no theorem would have. Once a claim made in this pass was withdrawn one section later.

Two further patterns are worth naming. Questions of the form *which X is it?* dissolved rather than resolved — which topological charge sources gravity (none can), and how the observer relates to spacetime (there is no spacetime) — and each time the development's own type signatures had already said so. And the same substitution error, reading a *scalar* scale where A4' supplies a *directional* one, appears in at least four separate places.

A formalisation makes the theorems reliable. It does not make the *prose around them* reliable, and most of what was corrected here lived in the prose — in what a true theorem was being quoted as showing.